LEF MODEL · CHAPTER 24

Chapter 24: The Ring as a Grinding Conveyor, the Signatures on Moons and Planets, and the Atacama Block Test

Based on Paper 12 — 12. The Rings: A ring breaks, grinds and returns (Parts E–M: the grinding conveyor, signatures on moons and parents, the Atacama rock and the 5 falsification criteria)

01 · THE MYSTERY

The mystery: how does a ring shape the surfaces of moons and planets — and why are there no rings around moons?

In Chapter 23 we saw that a planetary ring exists as an equilibrium flow that feeds on the parent and returns material to it when its day is shorter than 13.4 hours. In Parts E–M of Paper 12 the LEF model examines the morphological and physical traces the ring leaves behind:

1. **Why do Pan, Atlas and Daphnis have an equatorial "skirt" like a flying saucer — while Deimos was polished into a smooth ball (G.1–G.4)?** Although Deimos has stronger self-gravity than Pan and Atlas, Pan and Atlas accumulated a prominent equatorial ridge, while Deimos lost material and was worn down completely!

2. **Which rings in the solar system are doomed to disappear in the near future (F.3)?** Why do certain narrow rings at Uranus and Neptune survive for now with no shepherd moon and no feeding from the ground?

3. **Where on the surface of Mars can remnants of its ancient ring be found — and why was the "Atacama" rock block, drilled in April 2026, ruled out (H.2–H.3)?**
4. **Why has no ring been discovered around any moon in the solar system — and how do small bodies (Haumea, Chariklo, Quaoar) keep rings in a 1:3 resonance (J.1–J.2)?**

02 · AN IMAGE

An image: a one-way grinding conveyor, a sticky snowball against a grindstone, and stripes in Saturn's sky

Four tangible images explain the internal sorting in the ring and the signatures it imprints on its surroundings:

- **A one-way grinding conveyor where lumps go in and dust comes out (E.1–E.3):** inside the ring, a heavy particle and a light particle at the same radius move at the same Keplerian speed and so do not settle by density. The sorting happens **through the transfer of momentum in collisions**: a stiff lump ~ 100 m across keeps its stable orbit and clears clean lanes for itself in the shape of **"propellers"** in Saturn's ring A, while tiny particles receive random speeds and spread vertically around the thin layer (10 m) of the large lumps. The size distribution $n(D) \propto D^{-3}$ is a classic grinding signature: large lumps indicate a fresh entry into the ring!
- **A frost-coated snowball against a silicate grindstone (G.2–G.3):** the composition of the ring dictates whether a moon passing through it will *gain mass* or *be worn down*: cold, frost-coated ice grains stick to each other on slow contact (Bridges et al. 1984) and built the equatorial ridges on Pan, Atlas and Daphnis; while the ancient ring of Mars was made of condensed vapours and hard silicate grains that acted as **a grindstone**, polishing Deimos without adding mass to it!
- **Full immersion in water against a thin thread around a ball (the D/H ratio, G.4):** when the moon's diameter is smaller than the ring's thickness (D < H,

as with Deimos, 14–22 km, inside the thick vapour ring of Mars, or Methone), the whole surface is immersed and smoothed; when D > H (Pan, Atlas and Daphnis inside a disc tens of metres thick), the poles stick out above the ring and a wide equatorial belt is built; and when $D \gg H$ (Iapetus), the ring acts as a thin thread.

- **Two rain channels — neutral material at the equator and dark stripes along the magnetic field lines (H.1):** in Cassini's final dive in 2017, two channels of fall from the rings onto Saturn were measured: neutral molecules and nanometre silicate grains go straight down to the equator, while ionised water molecules flow along the magnetic field lines to the middle latitudes (25°–55° on both sides of the equator), neutralise H_3^+ ions and create dark stripes in infrared light!

03 · SCIENCE AND MODEL

Science and model: the life cycle of the ring, the signatures on moons and the Atacama block test (Parts E–J)

Parts E–J of Paper 12 present sharp observational tests to distinguish ring remnants from ordinary geological phenomena:

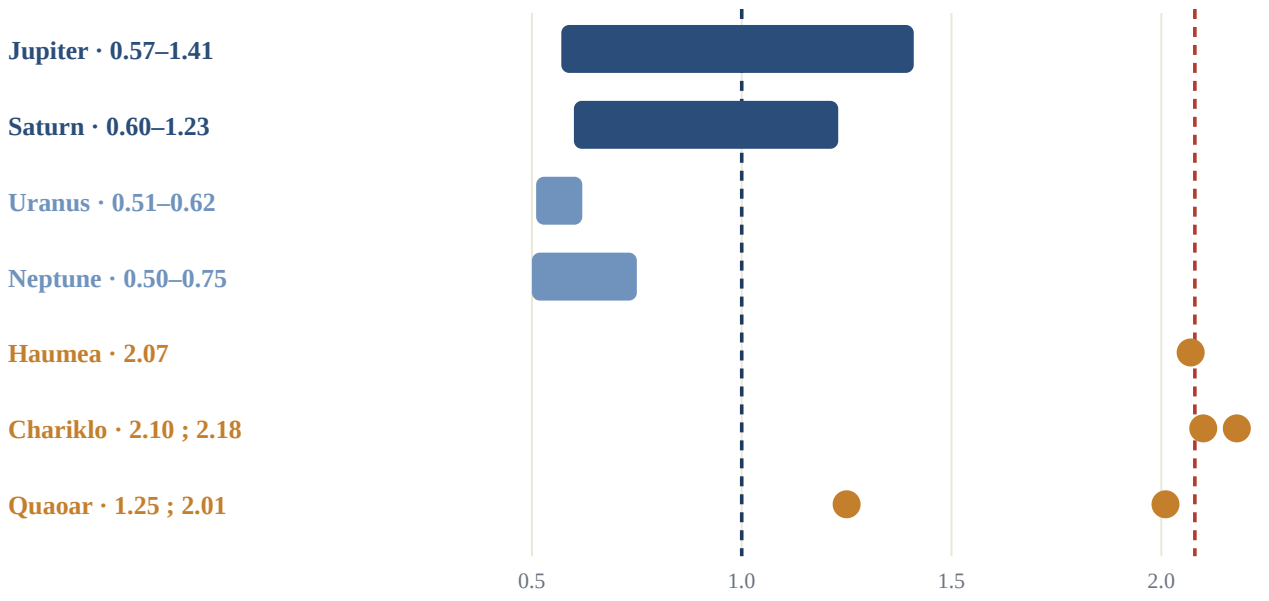
1. **The life cycle of the ring and the rings expected to disappear (F.1–F.3):** a ring that has no feeding moon and no leakage from the ground is in a stage of decay and is expected to disappear. So **the nine narrow moonless rings of Uranus** ($\zeta, 6, 5, 4, \alpha, \beta, \eta, \gamma, \delta$) and **Neptune's Galle and Lassell rings** are classified as dying rings left below the corotation line after the escape of water and the freezing, against Uranus' ε, ν, μ rings and Neptune's Adams and Le Verrier rings, which are held by moons.

2. **The morphological identification rule for a moon that passed through a ring (G.1–G.5):** a completely smooth silicate moon (without resurfacing geological activity as at Europa or Enceladus) indicates a long stay in the heart of a ring: on Deimos the median depth of the hollows is 1.9% of their width (84% shallower than 6%) against 5.5% on Phobos, and its orbit is

almost perfectly circular (e = 0.0003). The crossing rate depends on the moon's mass: a small moon migrates slowly and accumulates a deep signature, while a large moon is pushed fast and crosses the ring in a short time.

3. **Ruling out rings around moons and the stabilisation of small-body rings in a 1:3 resonance (J.1–J.2):** no moon in the solar system holds a ring (even the suspected ring around Rhea was completely ruled out in Tiscareno et al. 2010), because the gravity of the parent planet dominates and captures any material ejected from a moon into the parent's ring. By contrast, around small bodies (Haumea, Chariklo and Quaoar), four of the five rings sit above the corotation line in the range $2.01–2.18\,a_c$ — right at the radius of **the 1:3 spin–orbit resonance** ($a_{1:3} = 3^{2/3} a_c \approx 2.08\,a_c$), where a particle completes one orbit for every three rotations of the parent.

4. **The test of the "Atacama" rock block on Mars (25 April 2026, H.3):** when the Curiosity rover drilled into the detached rock block "Atacama" (~ 45 cm, ~ 13 kg) in Gale crater, the possibility was examined that it is a solid remnant that fell from Mars' ancient ring. The test ruled this out completely (a layered sedimentary rock of hydrated sulphate salts with desiccation cracks and a composition identical to the layer beneath it, belonging to the Mount Sharp unit) and set **four strict criteria for identifying a genuine ring remnant on Mars**: a volcanic/spongy rock with no water-laid layering, dry minerals that underwent heating, a fusion crust from atmospheric entry, and a composition foreign to the local geological layer (matching Phobos and Deimos).



Ring position in units of the corotation line · blue: the corotation line · red: the 1:3 resonance at 2.08

The two families of rings: the rings of the giants around and below the corotation line, and the rings of the small bodies above it, clustered around the 1:3 resonance.

`The effect of solar radiation on the rings — the "spokes", the ionised rain and Poynting–Robertson drag (Part I)`

Solar radiation acts on the rings by three mechanisms: (1) electrostatic charging of micrometre dust that floats above the plane of ring B as **"spokes"**, peaking around the equinoxes; (2) UV ionisation of water molecules, which speeds up **the ionised rain** along the magnetic field lines; and (3) **Poynting–Robertson radiation drag**, which brakes tiny dust particles and drops them back inward onto the parent!

04 · THE MECHANISM IN WORDS

The mechanism in words: the identification criteria on Mars, the systematic survey and the five falsification tests

Parts H.2–H.3 and K–M of Paper 12 gather the geological traces on the surfaces of the planets and the experimental rules by which the ring theory of the LEF model can be verified or falsified:

1. **The four search targets for remnants of Mars' ancient ring (§H.2):** when Mars' steam-and-silicate ring dried out 3.5–3.7 billion years ago, its material settled on the surface. The four search targets are: (a) thick, low-density equatorial deposits with no ground flow source (such as **the Medusae Fossae Formation**); (b) a deuterium–hydrogen (D/H) signature that departs from the local surroundings and matches Phobos and Deimos; (c) stones of volcanic composition and a dark, hard marker layer in Mount Sharp in Gale crater; and (d) symmetric sediment bands at middle latitudes if Mars had an active dipolar magnetic field while the ring existed.

2. **Summary of the systematic survey for each body (Part K):** each system in the solar system has a clear role in the ring cycle: the Earth (no lasting ring because the pear swept the equatorial band); Mars (an ancient ring from the ground that moved Deimos and Phobos and disappeared when the crust closed); Jupiter (a dust ring from moons only because of the thickest gas lid); Saturn (a giant ice ring in equilibrium flow from the ground and from moons); Uranus and Neptune (narrow, dark remnants below the corotation line after the escape of water, with the moonless rings dying); and Haumea, Chariklo and Quaoar (dense rings above the corotation line in a 1:3 resonance that do not coalesce into moons).

3. **The five sharp falsification criteria of Part L.3:** the model sets out five explicit falsification conditions: (1) direct observation of a moon coalescing out of a ring disc; (2) discovery of a completely smooth silicate moon that never stayed in a ring environment; (3) measurement of a planetary ring whose isotopic composition contradicts the parent and matches a source outside the system; (4) discovery of a massive ring far above the corotation line of a large planet with no stabilising orbital resonance; or (5) documentation of a ring with no feeding moon and no feeding from the ground that keeps its mass over geological time without being depleted.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The size distribution of lumps in a grinding ring and the radius of the 1:3 spin–orbit resonance that traps rings around small bodies are formulated as follows (§E.2, §J.2):

$$n(D) \propto D^{-3}$$

$$a_{1:3} = 3^{2/3}\, a_c \approx 2.08\, a_c$$

The power law $n(D) \propto D^{-3}$ characterises a system of constant crushing and grinding in collisions (Zebker et al. 1985), while the resonance radius $a_{1:3} \approx 2.08\, a_c$ explains why four of the five rings of Haumea, Chariklo and Quaoar sit exactly in the range 2.01–$2.18\, a_c$ above the corotation line.

06 · THE SIGNATURE

The signature in reality: the Atacama block test (April 2026) and the rings expected to disappear

Two central observational findings illustrate how the LEF ring theory stands up to a live empirical test (§F.3, §H.3):

- **The screening protocol of the Atacama rock at Curiosity (25 April 2026):** the geochemical analysis of the detached Atacama block (~ 45 cm, ~ 13 kg) in Gale crater showed that it is a sedimentary rock of hydrated sulphate salts with desiccation cracks, identical to the layer beneath it — and so the four threshold conditions (volcanic/spongy texture, dry minerals, a fusion crust and a foreign composition) were set for identifying a genuine ring remnant on Mars.
- **The list of dying rings in the solar system (§F.3):** the nine narrow moonless rings of Uranus ($\zeta, 6, 5, 4, \alpha, \beta, \eta, \gamma, \delta$) and Neptune's Galle and Lassell rings are expected to keep thinning and to disappear in future observations, while

Saturn's rings D, C and B will keep existing as long as the inner pump feeds them.

The lab box: the effect of the ring's composition and the D/H ratio on moons, the systematic survey and the falsification criteria (Paper 12, Parts E–M)

Moon	Diameter / density	Self-gravity (m/s^2)	Ring environment and D/H ratio	The morphological signature
Deimos	**12.4 km / 1,471 kg/m^3**	**0.0025**	**Silicate and vapours (D < H, full immersion)**	**Fully worn into a smooth ball (median hollow depth 1.9%)**
Phobos	22.2 km / 1,860 kg/m^3	0.0058	A fading gaseous edge	Only partial wear; kept its grooves (5.5%)
Pan, Atlas and Daphnis	**7.6–30 km / 340–460 kg/m^3**	**0.0004–0.0019**	**Water ice (D > H, the poles stick out)**	**Accumulation of a prominent equatorial ridge ("saucer" shape)**
Methone	2.9 km / 310 kg/m^3	0.0001	Ice in an arc near ring G (D < H)	Smooth and coated

System in the solar system	State of the ring today	Feeding source	Position relative to the corotation line and the LEF interpretation
Earth / Mars	None / existed in the past	Mars: steam and rock from the bulge	Earth: the pear swept the line; Mars: disappeared 3.5–3.7 Ga ago when the crust closed
Jupiter	Thin dust	Inner moons (Metis to Thebe)	Around the line ($0.57–1.41\ a_c$); the thick gas lid seals inner leakage
Saturn	**Massive (95%–99% ice)**	**From the ground (D, C, B, A) and from moons (E, F, G)**	**A constant equilibrium flow (P = 10.6 h < 13.4 h); edge of A at the Roche limit**
Uranus / Neptune	Narrow and dark (and arcs)	Some from moons, some moonless	Below the line ($0.50–0.75\ a_c$); the moonless rings are expected to disappear

Haumea, Chariklo, Quaoar	**Dense**	**Haumea: the ends of the long axis**	**Above the line (1.25–$2.18\ a_c$), trapped in the 1:3 resonance; they do not build moons**

07 · NEXT

The door ahead

In Chapters 23–24 we completed the ring theory of the LEF model: we derived **the universal threshold day of $P^* \approx 13.4$ hours**, explained Saturn's rings as a constant flow in dynamic equilibrium, followed the fate of the water and the angular momentum of Uranus and Neptune, and defined the morphological signatures on moons (the wearing of Deimos against the ridge-building of Pan and Atlas) and on Mars (the Atacama rock test and the Medusae Fossae deposits).

Chapter 25

The Fundamental Requirements of the LEF Model (R1–R8) — Isotopes, the Core, the Crust, the Centre-of-Mass Offset and the Exposure Zones →

← Chapter 23 · Prologue

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